

CHAPTER THREE

INFLUENCE OF WASTEWATER CHARACTERISTICS ON PROCESS DESIGN

by

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1. INTRODUCTION

In this chapter considerations in the design of nutrient removal activated sludge processes are qualitatively described to provide a background for the subsequent chapters in which the process design for COD removal only (Chapter 4), nitrification (Chapter 5), nitrification-denitrification (Chapter 6) and biological phosphorus removal (Chapter 7) are quantitatively set out – in these chapters detailed equations, design charts, sample calculations and recommendations for design are presented.

2. RELATIVE IMPORTANCE OF CHARACTERISTICS

In the design of activated sludge processes the characteristics of the wastewater are of prime importance. These characteristics govern both the selection of the process and the removals of nitrogen and phosphorus attainable in the process.

Influent wastewater characteristics of crucial importance are:

- (1) Mean influent COD and TKN concentrations (S_{ti} and N_{ti} respectively) and mean daily flow (Q)
- (2) Influent readily biodegradable COD fraction (f_{bs})
- (3) Influent TKN/COD concentration ratio (N_{ti}/S_{ti})
- (4) Influent (Total P)/COD concentration ratio (P_{ti}/S_{ti})
- (5) Maximum specific growth rate of the nitrifiers at the reference temperature of 20°C (μ_{nm20})
- (6) Average minimum and maximum temperatures (T_{min} and T_{max}) in the process.

Influent wastewater characteristics of lesser importance are:

- (7) Unbiodegradable soluble COD fraction (f_{us})
- (8) Unbiodegradable particulate COD fraction (f_{up}).

Influent characteristics normally not important in design are:

- (9) Unbiodegradable soluble organic nitrogen fraction (f_{nu}) and
- (10) Ammonia fraction of the TKN (f_{na}).

3. EFFECTS OF CHARACTERISTICS ON DESIGN

3.1 Influent COD and influent flow

The mean influent COD concentration, S_{ti} ,* and mean flow per day, Q , define the mass COD load day on the plant, $M(S_{ti})$, by

$$M(S_{ti}) = S_{ti} \cdot Q$$

This COD mass consists of various fractions, as set out in Chapter 2, and these fractions are different for un-settled and settled municipal waste flows. However, to indicate the effect of the COD on the plant design, accept a specific waste flow influent. Then, for a specified sludge age, once $M(S_{ti})$ is known, theoretically the mass of sludge in the reactor, $M(X_v)$ or $M(X_t)$, and the mass of oxygen required daily for carbonaceous degradation, $M(O_c)$, also are known. The sludge age must be specified as the mass is very sensitive to the sludge age. Knowing the mass of sludge, the volume of the reactor is found from the specified mixed liquor concentration, X_v or X_t , i.e. the mass of sludge is "diluted" to the required concentration by providing the appropriate volume,

$$V_p = M(X_v)/X_v \text{ or } M(X_t)/X_t$$

Knowing the volume, one can now calculate, if so desired (it is not essential) the nominal hydraulic retention time, R_{hn} , of the reactor,

$$R_{hn} = V_p/Q$$

From a process design point of view the hydraulic retention time is of little interest: For two waste flows, having the same COD fractional constitution, provided $M(S_{ti})$ and X_v are the same for both, the volumes of the two plants and the oxygen demand per day will be identical irrespective of whether the first flow consists of a low S_{ti} and high Q and the second of a high S_{ti} and low Q . The nominal hydraulic retention time will differ, the first plant having a short and the second a long retention time respectively but this does not affect the degradation action, except insignificantly. The influent flow is important in the design of the settling tanks – for the same upflow velocity (overflow rate) in the secondary

*The mean influent COD, S_{ti} , is the *weighted* mean concentration; it is *not* the mean of the COD concentrations measured over the day on the influent. The magnitude of S_{ti} is found from measuring every hour, say, the COD, $S_{ti}(t)$, and flow, $Q(t)$, and forming

$$S_{ti} = \left[\sum_{t=1}^{24} S_{ti}(t) \cdot Q(t) \right] / \left[\sum_{t=1}^{24} Q(t) \right]$$

If the wastewater is settled, S_{ti} refers to the measurements on the effluent from the primary settling tank.

settling tank, the size for the first flow (high) will be larger than that for the second flow (low). Thus knowing only the COD load (and its fractional components), and the sludge age, the volume of the plant and the mean carbonaceous oxygen requirements can be estimated rapidly, also, the mass of sludge to be wasted per day. Knowing the mean flow, the size of the settling tanks can be estimated. Once these basic estimates are available, the effects of cyclicity in the flow and load on the peak carbonaceous oxygen demand and peak flow through the settling tank, then can be superimposed.

It was stated above that the mass of sludge, $M(X_v)$, depends on $M(S_{ti})$ and the sludge age. As the sludge age increases so the $M(X_v)$ increases likewise, the mass being roughly proportional to the sludge age; this is illustrated in Figure 3.1 where the mass of sludge in the process per unit influent COD, for unsettled and settled "normal" municipal waste flows, are shown plotted versus sludge age. Consequently the longer the sludge age, the bigger the reactor volume required to deal with the same $M(S_{ti})$. In Figure 3.1 are also shown plotted the carbonaceous oxygen demands per unit COD input for settled and unsettled waste flows. As for $M(X_v)$ the respective magnitudes of $M(O_c)$ do not differ greatly between settled and unsettled wastes. The difference arises principally from the fraction of unbiodegradable particulate COD, f_{up} , in the two wastes being higher in unsettled than in settled waste. However, the total masses of sludge and carbonaceous oxygen demands for the two flows will differ appreciably because approximately 40% of the COD is removed in the primary settling tanks.

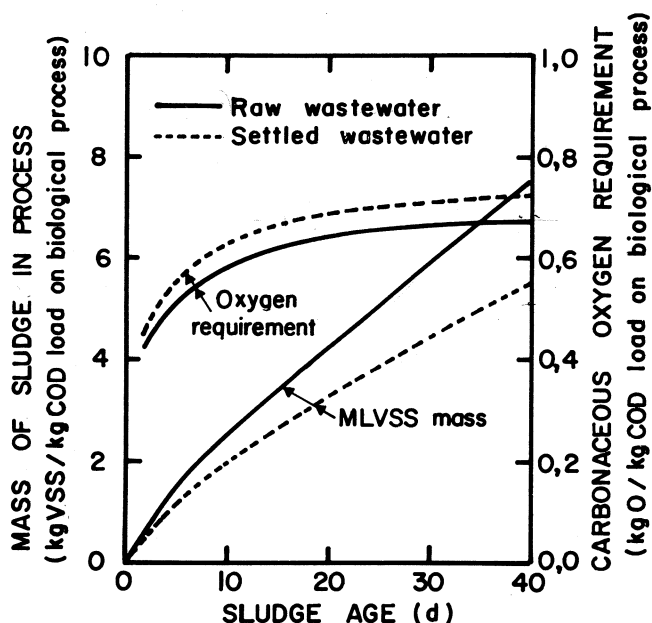


FIGURE 3.1: Mass of volatile solids and mass of carbonaceous oxygen demand per day per kg COD load per day versus sludge age for settled and unsettled municipal waste flows.

3.2 Influent TKN, μ_{nmT} and temperature

In the discussion above, to calculate the sludge mass it was stated that the sludge age must be specified. Selection of the sludge age probably is the most important decision the designer must make, particularly when the objective includes nutrient removal. If removal of carbonaceous material only is required, the sludge age can be quite short; three days will be enough for COD removal – the sludge age selected is dependent more on the expected settleability of the sludge than on the COD removal. If nitrification only is required the sludge age must be sufficiently long to allow the nitrifying organisms to grow and metabolize virtually all the available nitrogen. If denitrification is required in addition to nitrification, a fraction of the sludge mass must be kept unaerated to denitrify the nitrate generated. As the nitrifiers are obligate aerobes, this implies that in nitrification-denitrification plants the sludge age must be increased above that where nitrification only is required, to ensure that the aerobic sludge age is still adequate for nitrification.

The maximum specific growth rate constant of the nitrifiers, μ_{nm} , defines the minimum aerobic sludge age for the initiation of nitrification. This minimum sludge age must be increased by a further 25 per cent to ensure that nitrification is near complete. Now μ_{nm} tends to be specific to the waste flow and can range from 0,20 to 0,70/day at 20°C. Also μ_{nm} is very sensitive to temperature, the value halving for every 6°C drop in temperature. The "aerobic" sludge age is given by $1/\mu_{nm}$; it is clear therefore that, in a nitrification plant, a low μ_{nm} value for the waste flow coupled with a low temperature during the winter, will require a long sludge age to ensure efficient nitrification throughout the year. In nitrification-denitrification plants the process sludge age needs to be increased further to accommodate the effects of the unaerated fraction of the sludge mass. These factors in combination can cause that whereas a sludge age of 4 days for a $\mu_{nm} = 0,3$ at 20°C will be adequate theoretically for a purely nitrifying aerobic plant, for a nitrifying-denitrifying plant with $\mu_{nm20} = 0,3$, $T_{min} = 12^\circ\text{C}$ and an unaerated sludge mass fraction of 0,5, the minimum process sludge age will be about 15-20 days to ensure a high efficiency of nitrification.

Provided the sludge age is adequate, the mass of nitrate formed by a process is given by the mass of TKN in the influent minus the mass of nitrogen incorporated in the sludge mass wasted every day. This gives the *nitrification capacity* of the plant (usually expressed as nitrate formed/unit influent flow). The higher the TKN concentration in the influent so, in general, the higher the nitrification capacity – we can form the rough rule that the nitrification capacity is proportional to the TKN concentration in the influent.

3.3 Readily biodegradable COD, slowly biodegradable COD

3.3.1 Nitrogen removal

In the previous section it was mentioned that a fraction of the process sludge mass must be unaerated to

denitrify the nitrate generated in the aerobic fraction, but the magnitude of this unaerated fraction was still undetermined. It is the intention now to describe qualitatively how this unaerated fraction is determined and how the influent characteristics affect the determination.

In the absence of oxygen, nitrate serves as an electron acceptor. With oxygen, the process designer has the problem of determining the oxygen demand for known COD load; in denitrification the problem is to determine the COD to denitrify a known mass of nitrate. The COD for denitrification comes from three sources (1) influent readily biodegradable soluble COD, (2) influent slowly biodegradable particulate COD and (3) particulate COD derived from the death of organisms. Usually (2) and (3) are lumped together in formulating the reduction of nitrate.

In the nitrification-denitrification process because the same sludge mass generates the nitrate and denitrifies it (called the single sludge process), the sludge mass normally is subdivided into two reactors in series, the one aerated (to nitrify) and the other unaerated (to denitrify).^{*} Two basic configurations have been developed, the Modified Ludzack-Ettinger (MLE) configuration, Figure 3.2, and the Wuhrmann configuration, Figure 3.3. In the MLE configuration, the

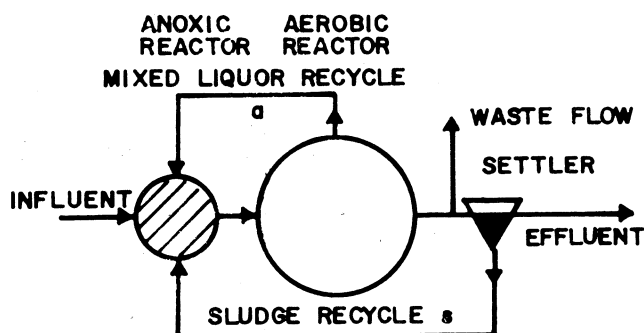


FIGURE 3.2: Modified Ludzack-Ettinger nitrification-denitrification activated sludge process.

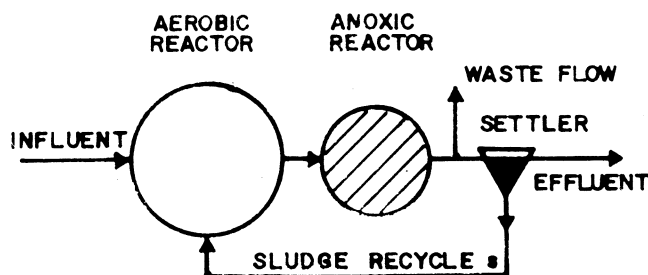


FIGURE 3.3: Wuhrmann nitrification-denitrification activated sludge process.

first reactor is left unaerated, the second is aerated. The nitrate generated in the second reactor is transferred to the first via the underflow and an inter-reactor (internal) recycle. In the Wuhrmann configuration the nitrate is generated in the first reactor and transferred by the normal serial flow to the second unaerated reactor – an internal recycle is not required, the only recycle being the underflow recycle to transfer the sludge from the secondary settling tank back to the first reactor.

The denitrification action can best be explained by considering the unaerated reactors as plug flow reactors. (This can be shown to provide valid explanation also for completely mixed reactors). In the unaerated reactor of the MLE configuration (called the primary anoxic reactor), the denitrification takes place at two simultaneously occurring rates, the first due to the influent readily biodegradable COD, the second due to the particulate COD derived principally from the influent, see Figure 3.4a. The first rate is very fast, the reaction being complete in 6 to 20 minutes and it utilizes all the readily biodegradable COD. The second rate is slower, at about 1/7 of the first rate, and continues for the total time in the reactor. In the unaerated reactor of the Wuhrmann configuration, (called the secondary anoxic reactor), a single denitrification rate is observed, due principally to the slowly biodegradable particulate COD derived from death of the organisms, (Figure 3.4b). This source of COD becomes available at a low rate and denitrification rate accordingly is low, about 2/3 that due to particulate COD in the MLE primary anoxic reactor.

The nitrate removal can be expressed either as total mass removed per day or as mass removed per litre of flow, i.e. as a concentration relative to the flow $\text{mg}\Delta(\text{NO}_3\text{-N})/\text{l flow}$. The latter method is the most convenient because it allows ready comparison with the TKN concentration in the influent flow, $\text{mgTKN}/\text{l flow}$. We have stated that the removal is due to the complete utilization of the readily biodegradable COD and the partial utilization of the biodegradable particulate COD. The denitrification achievable due to the latter can be formulated in terms of a denitrification rate constant, K , the *active* mass of sludge generated per litre of influent and the fractional mass of the process sludge contained in the specific anoxic reactor, as follows: (See Chapter 6, Section 4).

In the primary anoxic reactor

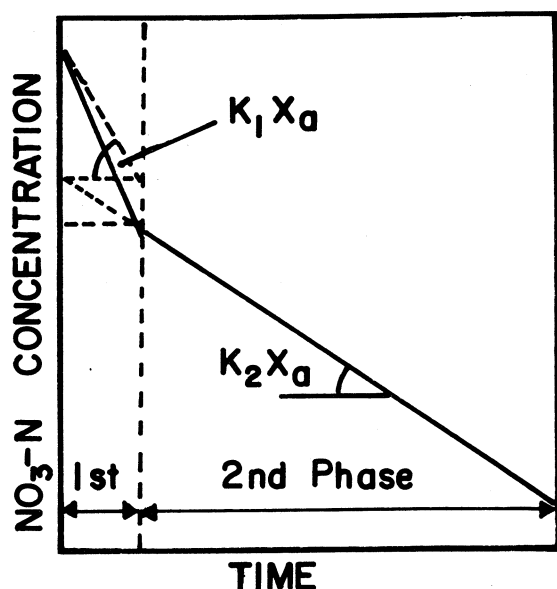
$$\Delta N_1 = C \times (\text{mass of readily biodegradable COD per litre of influent}) + K_2 \times (\text{mass of active sludge generated by one litre of influent}) \times (\text{mass fraction of sludge in the primary anoxic reactor}) \quad (3.1)$$

In the secondary anoxic reactor

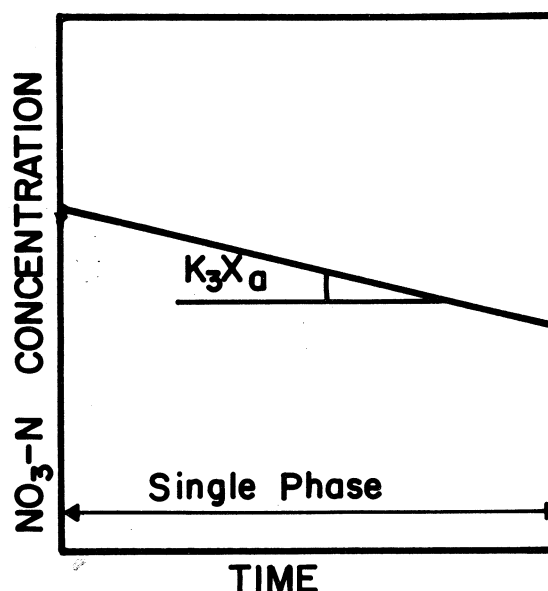
$$\Delta N_3 = K_3 \times (\text{mass of active sludge generated per litre of influent}) \times (\text{mass fraction of sludge in the secondary anoxic reactor}) \quad (3.2)$$

The values of K_2 and K_3 have been determined experimentally and found to remain virtually constant for sludge ages between 10 and 30 days, at a fixed

^{*} Other processes, in which the sludge is sequentially aerated and left unaerated, also have been proposed and built, either sequentially in time in a single completely mixed reactor or sequentially in space as, for example, in the Carrousel system.



(a) Primary anoxic



(b) Secondary anoxic

FIGURE 3.4: Denitrification time behaviour in plug flow anoxic reactor, (a) primary anoxic and (b) secondary anoxic reactors.

temperature. This is also indicated by simulation using the general activated sludge model of Dold, Ekama and Marais (1980) as extended to include denitrification by van Haandel, Ekama and Marais (1981). The two equations developed above, expressed in technical terms are for primary and secondary anoxic reactors respectively

$$\Delta N_1 = f_{bs} (1 - f_{up} - f_{us}) S_{ti} / 8.6 + K_2 \frac{Y_h R_s}{1 + b_h R_s} S_{ti} (1 - f_{us} - f_{up}) \frac{R_{hn1}}{R_{hn}} \quad (3.3)^*$$

$$\Delta N_3 = K_3 \frac{Y_h R_s}{1 + b_h R_s} S_{ti} (1 - f_{us} - f_{up}) \frac{R_{hn3}}{R_{hn}} \quad (3.4)^*$$

R_{hn1}/R_{hn} and R_{hn3}/R_{hn} are the ratios of the nominal retention times of the primary and secondary anoxic reactors respectively with respect to the total. The ratios respectively are equal to the primary and secondary anoxic sludge mass fractions. Here as for aerobic biological degradation, the nominal retention is not the basic parameter even though one may formulate the equations to include it. From Eqs. 3.3 and 3.4 the denitrification potential is a function of the influent COD, S_{ti} , but is approximately so only, being modified by the magnitude of the unbiodegradable fractions of the influent COD, f_{us} and f_{up} .

In Eq. 3.1, all the readily biodegradable COD is utilized; the denitrification attained from this source in the influent is very effective and constitutes about 50% denitrification achieved in "normal" municipal wastewater. The removal due to the particulate biodegradable COD typified by K_2 and K_3 in Eqs. 3.1 and 3.2 *inter alia*, is directly proportional to the anoxic sludge mass fractions of the respective anoxic reactors (i.e. f_{x1} and f_{x3}).

These fractions cannot be increased *ad lib*; for a fixed sludge age the magnitude of the sum of these fractions is subject to the proviso that efficient nitrification must be maintained and this in turn is fixed by the aerobic sludge age. In this way there is an upper limit to the sum of anoxic sludge mass fractions which limits the nitrate removal achievable. Furthermore because K_3 is roughly 2/3 of K_2 , the nitrate removal due to K_3 in the secondary anoxic reactor per unit volume is only about 2/3 as effective as that in the primary anoxic reactor so that per unit sludge mass fraction the primary anoxic reactor is more efficient than the secondary anoxic reactor.

The nitrate removals, ΔN_1 and ΔN_3 , are called the *denitrification potentials of the respective reactors*. The term *potential* is used because in the primary anoxic reactor (as in an MLE process), for example, the denitrification potential can be achieved only if sufficient nitrate, equal to the potential, is recycled to this reactor – if less is recycled the full potential of the reactor is not realized and the *denitrification performance* (or *denitrification capacity*) is less than the denitrification potential. When the performance is less than the potential then the performance can be increased by increasing the recycle. Now the recycle contains dissolved oxygen in addition to nitrate, and the oxygen utilizes the COD preferentially; in consequence, recycle ratios in excess of 4 to 6 usually do not significantly improve the performance, but even can cause a *decrease* in performance due to the high mass of DO transported to the anoxic reactor via the recycle flow. If more nitrate is recycled than that defined by the potential, only the potential value is removed. The excess is returned from the anoxic reactor to the aerobic reactor; in such a situation the recycle can be reduced appropriately without affecting the nitrogen removal by the process.

The combined primary and secondary anoxic sludge mass fraction i.e. the total unaerated sludge

*These equations are for illustrative purposes only, the meaning of the symbols can be found in Chapters 4 to 6.

mass fraction, cannot be increased *ad lib.*, for two reasons:

- (1) Nitrification requires a minimum aerobic sludge age and to preserve this, as the unaerated mass fraction is increased so also the process sludge age must be increased to preserve the minimum aerobic sludge age; increasing the sludge age, however, increases the total mass of sludge and hence the total volume of the reactor system. Furthermore, any increase has a relatively small effect on nitrate removal because the increase contributes to denitrification only through the K_2 or K_3 components and these are relatively inefficient. Practically, for South African conditions, the upper limit to a sludge age appears to be about 30 days.
- (2) When the nitrification rate is high the aerobic sludge age needed is small so that with a long sludge age it would be possible to have a large unaerated sludge mass fraction. Experimental data, supported by simulation studies, show that for sludge ages between 20 and 30 days, if the unaerated fraction of a process increases above about 70% at 20°C and about 50% at 12°C the time available for aerobic degradation is insufficient to process all the particulate influent biodegradable COD and the process acquires a degradation pattern still ill understood. Consequently for design under South African conditions for sludge ages of 20 to 30 days, the unaerated mass fraction should not exceed about 55 to 60%.

We have noted earlier that the denitrification potentials are approximately proportional to the influent COD concentration, S_{ti} . Now previously it was pointed out that the nitrification capacity is approximately proportional to the influent TKN. Consequently the TKN/COD ratio of the influent is a good parameter in terms of which to assess process nitrification-denitrification behaviour because it gives approximately the ratio of the nitrate generated to the denitrification potential. Now accepting that

- μ_{nm} will vary around 0,4 per day,
- the sludge age is in the range 20 to 30 days,
- "normal" municipal wastewater flows contain readily biodegradable COD fractions of about 20% with respect to the total influent COD,
- the internal recycle ratio is limited to a maximum of 6:1, and
- the minimum winter temperature is 14°C,

it would appear that for these conditions, if the influent TKN/COD is greater than about 0,10, the denitrification potential is such that it is not possible to denitrify all the nitrate generated even if all the unaerated fraction of the process is located in the primary anoxic reactor. Consequently in this situation the MLE process is the most efficient. If the influent TKN/COD ratio is less than about 0,09 then complete denitrification is possible.

Even if complete denitrification is possible this cannot be accomplished by using an MLE process only for the following reason:

Say the steady state nitrate concentration in the aerobic reactor is $N \text{ mg/l}$; if an internal recycle ratio of a is imposed and an underflow recycle ratio of s then $(a+s)/(a+s+1)$ of this nitrate is recycled and $1/(a+s+1)$ goes out with the effluent. To deal with the $1/(a+s+1)$ fraction a secondary anoxic reactor needs to be incorporated also in the process, to give the Bardenpho process first proposed by Barnard, 1973 (Figure 3.5). Of course the secondary anoxic reactor per unit volume is not as efficient as the primary because $K_3 < K_2$ so that the denitrification potential for the same unaerated total mass fraction will be lower than for an MLE process only, but this must be sacrificed and is the reason why complete denitrification is achievable only for a $\text{TKN/COD} \leq 0,09$ instead of 0,10. This significant reduction in the upper limit of the TKN/COD is principally because the a -recycle should not be increased above $a=6$ (to limit the feedback of oxygen to the primary anoxic reactor). If the oxygen in the recycle could have been reduced to zero then very high a -recycles could be utilized and it might not be necessary to employ a secondary anoxic reactor.

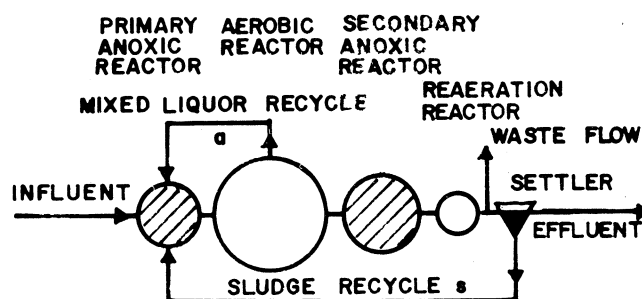


FIGURE 3.5: Bardenpho process for biological nitrogen removal – a combination of the modified Ludzack-Ettinger and Wuhmann processes.

As the TKN/COD ratio decreases below 0,09 so the size of the primary anoxic reactor may be decreased and the second anoxic reactor increased in order to maintain a low recycle, illustrated in Figure 3.6.

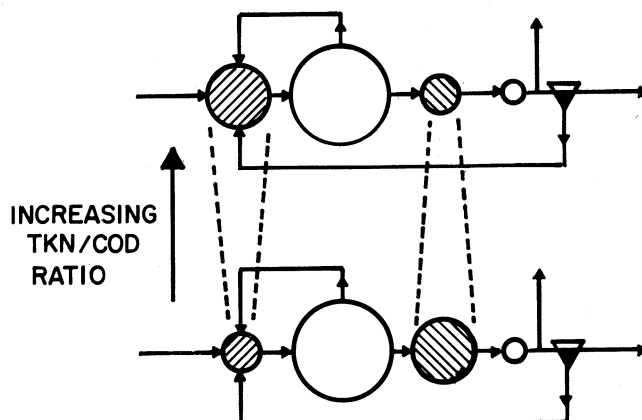


FIGURE 3.6: Effect of TKN/COD ratio on relative sizes of primary and secondary anoxic reactors in the Bardenpho process.

3.3.2 Excess phosphorus removal

There is considerable support for the view that under certain conditions some species of organisms, for example, the *Acinetobacter spp.*, can take up phosphorus in excess of its metabolic requirements. Some of the acinetobacter species are obligate aerobes, others can utilize nitrate as electron acceptor. The species is slow growing with rather specific requirements to the type of nutrient it can utilize; it can utilize, for example, many of the lower fatty acids, but not sugars, such as glucose. Because it is slow growing, in aerobic systems, with mixed cultures of organisms, its growth is suppressed by faster growing organisms, because the faster growing organisms utilize most of the substrate and eventually dominates over the slow growing ones. This is the reason why it is not easy to find evidence of acinetobacter in purely aerobic processes.

From the discussion above, how then does it happen that the poly-P organism proliferates in some systems containing unaerated fractions? Assume that the organism already contains some excess phosphate. This phosphate is present as the radical PO_4^- but bound together to form polyphosphate chains. The bonds between the radicals in the chain contain appreciable energy. If the acinetobacter is discharged to an anaerobic reactor (i.e. a reactor that has neither nitrate nor oxygen present), in which substrate is present suitable for growth of *Acinetobacter spp.*, it breaks down some of the bonds in the polyphosphate chain, releasing phosphate radicals to the surrounding liquid. The bond energy thus released is utilized to absorb, complex and store the substrate in the organism – we say it sequesters the substrate. In contrast, in the anaerobic zone, because no external electron acceptor (oxygen or nitrate) is available, the non poly-P organisms cannot utilize the substrate. On leaving the anaerobic zone and entering an anoxic zone (with nitrate present) or an aerobic zone (with oxygen present), the poly-P organisms now has electron acceptors available and metabolizes the stored substrate sequestered for itself in the anaerobic zone i.e. it can grow without being in competition with the non poly-P organisms for the balance of the energy still available in the surrounding medium.

In the anoxic and aerobic zones the poly-P organisms, beside using the sequestered substrate for growth, also use it to take up phosphate radicals (PO_4^-) from the surrounding medium, again to form phosphate chains for use subsequently when returned to the anaerobic zone. Thus, by having the propensity to store polyphosphate the acinetobacter and other poly-P organisms have a positive advantage over the non poly-P organisms to obtain energy for themselves in the apparently inhospitable anaerobic reactor. This explains the apparently anomalous growth response of the poly-P organisms.

The behaviour of the poly-P organisms in the anaerobic reactor is not yet well understood but a summary of the behaviour appears to be as follows: When a substrate is added that can be utilized *directly* by the poly-P organisms, such as acetate or butyric acid, the release of P (to obtain energy for sequestering the substrate) is very rapid, and the release is proportional

to the mass of substrate added. Also, the net removal of P from the anaerobic-anoxic-aerobic system increases as the substrate added increases, but insufficient data has been collected to express these responses quantitatively.

When sewage is added to the anaerobic reactor the behaviour differs in two aspects from that when acetate is added: (1) only a fraction of the COD added is implicated in the P release and P removal; (2) the rate of P release is significantly slower with sewage than with acetate addition and the release is of a different kinetic order.

- (1) The influent COD that induces P release seems to be associated with the readily biodegradable COD fraction. It has been observed consistently in laboratory scale processes that the release and removal of P increases and decreases as this fraction increases and decreases (see Figure 3.7) although it is not yet clear, and highly unlikely, that all the readily biodegradable COD is implicated in this action.

With regard to the particulate biodegradable COD, this fraction appears to have little effect on P release. This is not an unreasonable conclusion because the particulate COD requires adsorption, storage and extracellular breakdown prior to absorption by the organisms; even under aerobic conditions the rate of metabolism of the particulate COD is only at 1/7 to 1/10 the rate at which readily biodegradable COD is metabolized, Dold, Ekama and Marais (1981).

- (2) The slower rate of P release with sewage addition compared to that with acetate addition seems to indicate that the substrate suitable for P release becomes available indirectly to the poly-P organisms. This is further supported by the observation that whereas with acetate addition the P release appears to be zero order with respect to acetate concentration, (Figure 3.8), with sewage addition the release appears to conform to a first order reaction; in a plug flow anaerobic reactor receiving sewage the rate of release is high initially but dies away steadily along the reactor until it is virtually zero. This is also indicated in a series system of anaerobic reactors (Figure 3.9). The first order nature of the reaction in the series system is verified by plotting the log (release/reactor) versus the cumulative retention time, to give a linear plot, (Figure 3.10). Comparison between the release in a plug flow reactor and a completely mixed reactor also shows that for the same anaerobic mass fraction if the mass fraction is so small that all the P is not released in the plug flow reactor, the release in the completely mixed reactor is less than in the plug flow one, a behavioural pattern in conformity to that expected where a first order reaction controls.

According to the observed behaviour described above, at sludge ages about 20 to 25 days, the anaerobic mass fraction of a completely mixed reactor should be about 0.2 to ensure that 90% or

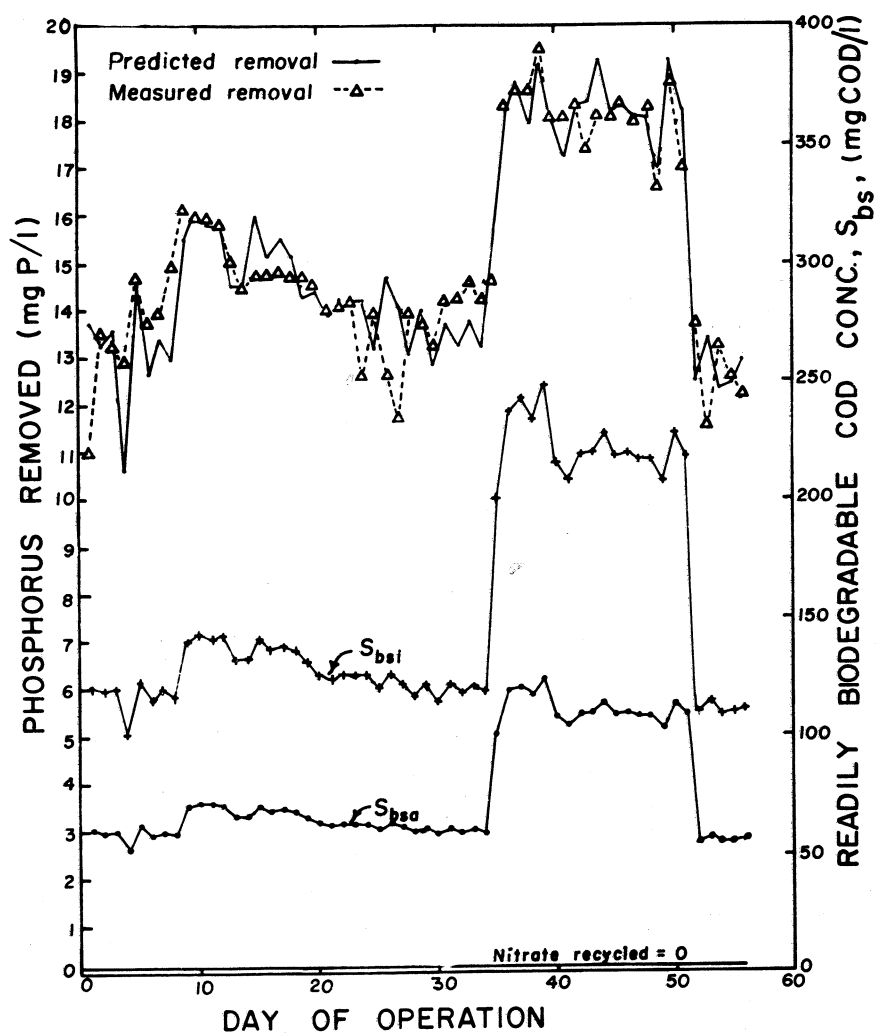


FIGURE 3.7: Experimental data showing the strong dependence of excess P removal (ΔP) on readily biodegradable COD in the influent (S_{bsi}).

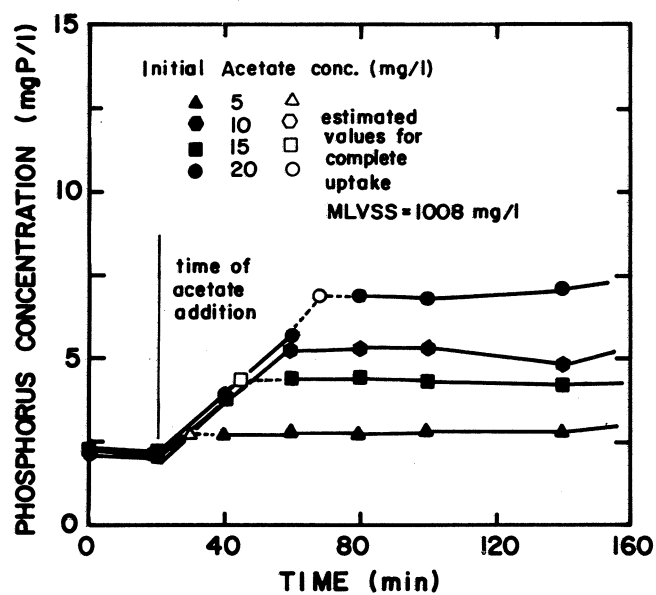


FIGURE 3.8: Phosphorus release in a batch anaerobic reactor with different concentrations of acetate added. Note the magnitude of the release is proportional to mass of acetate added but the rate of release is independent of the acetate concentration.

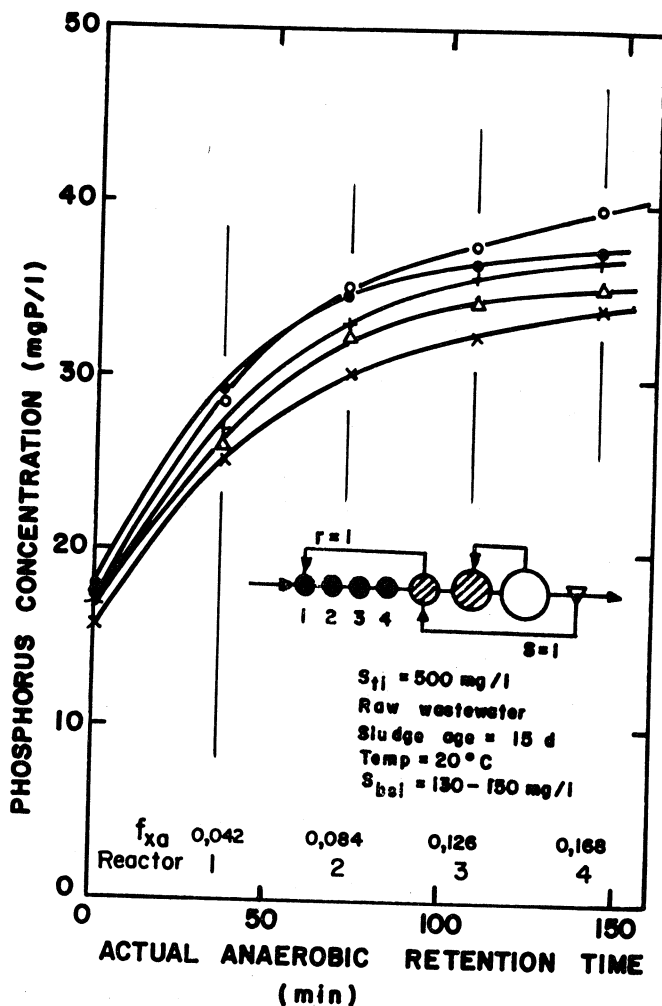


FIGURE 3.9: Phosphorus concentration in a four in series anaerobic reactor system, in a modified UCT process, receiving unsettled municipal sewage influent.

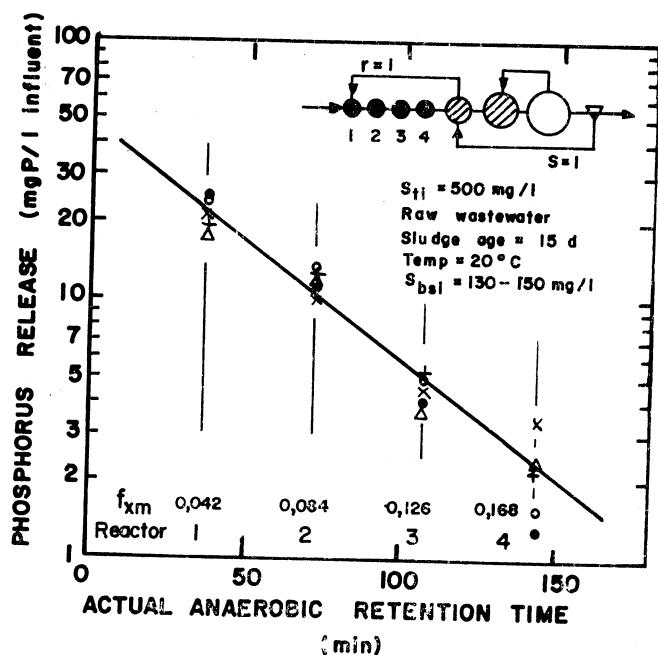


FIGURE 3.10: Phosphorus release per reactor in a four in series anaerobic reactor system in a modified UCT process receiving unsettled municipal sewage influent (see also Figure 3.9)

more of the P is released; if it were a plug flow reactor the mass fraction needs to be about 0.15 to give the same percentage release. Because the release rate is first order there is little to be gained by making the anaerobic mass fraction greater than about 20%. Mass fraction of 10% or less again will be too small allowing only about 50 to 60% of the potential release to take place. Then again, subdividing the anaerobic reactor into say two reactors in series gives improved usage of the anaerobic reactor mass fraction and requires a mass fraction of about 15% to give the same release as a completely mixed reactor of 20%. These fractions apply only for sludge ages of 20 days and longer.

The reasons for the different responses observed with acetate and sewage readily biodegradable COD additions are not yet clear but the following explanation could be valid: Most of the fractions of readily biodegradable COD, apparently, are not in a chemical form suitable for utilization by the poly-P organisms. In the anaerobic zone the non poly-P organisms, for *survival*, because no external electron acceptors are available, partially utilize the readily biodegradable COD, via the Embden-Meyerhof or other equivalent pathways, to generate a small amount of energy, by creating internal or endogenous electron acceptors: The outcome of this action is a breakdown of the original COD molecules to lower fatty acids and other similar forms. These are rejected by the organism and become available to the poly-P organisms for sequestration. Thus, where substrate is present in a form unavailable to the poly-P organism, the presence of non poly-P organisms can modify some of the COD fractions to forms suitable for use by the poly-P organisms. The kinetics of this transformation will govern the kinetics of sequestration by the poly-P organisms because the COD breakdown is the rate limiting step. The non poly-P organisms are, in terms of this hypothesis, essential to the growth of the poly-P ones.

3.3.3 Implication on process development

The rationalization of the poly-P behaviour leads to a number of conclusions that are of significance to designers of P removal processes:

- (1) In terms of the hypothesis set out above, the generation of the lower fatty acids by acid digestion of primary settling tank underflow, as proposed and implemented by the Municipality of Johannesburg, makes good sense and merits serious research attention.
- (2) Any external electron acceptor entering the anaerobic reactor, like nitrate or oxygen, will be utilized immediately by the non poly-P organisms acting on the readily biodegradable COD and thereby reduce the COD concentration available for the generation of substrate suitable for sequestration by the poly-P organisms.
- (3) The fact that the whole sequestration action is slow means that adequate anaerobic mass must be

made available to process as much as possible of the readily biodegradable COD source. However, because the reaction is first order it also implies that there is little merit in enlarging the anaerobic mass *ad lib.*, each increase giving a decreasing return on the previous increase. Also, series anaerobic reactors are more efficient and hence smaller anaerobic mass is necessary, to achieve the same effect than with a single completely mixed anaerobic reactor.

With regard to design implications, the basic requirement in a design is to provide an anaerobic zone which receives the influent containing the readily biodegradable COD fraction and so becomes available, directly or indirectly, to the poly-P organisms for sequestration. This zone must be large enough to allow the sequestration action to be as near complete as possible in order to obtain maximum effect. Furthermore, the zone should not receive any form of the external electron acceptor (nitrate, nitrite or dissolved oxygen) as these cause the fast growing non poly-P organisms to utilize the substrate before it can be sequestered by the poly-P organisms.

The first explicit design of a configuration, in which an anaerobic reactor is proposed, was due to Barnard* (1976). He accepted the Bardenpho process (Figure 3.5) as the basic configuration to nitrify and denitrify. A fraction of the primary anoxic reactor was isolated to serve as an anaerobic reactor. This reactor was placed at the head of the process and received the underflow recycle and the influent flow (see Figure 3.11). The reasoning behind this configuration was that the Bardenpho section of the process would give complete nitrification and denitrification so that the underflow would contain little or no nitrate, in this fashion allowing optimal conditions in the anaerobic reactor and maximum release of P. The process configuration has been variously called the Modified Bardenpho, Phoredox or 5 stage Phoredox process.

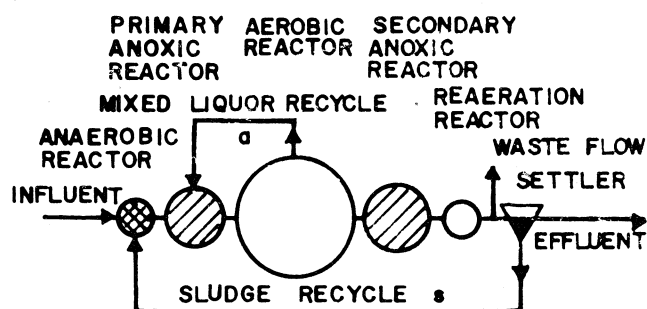


FIGURE 3.11: Phoredox (or modified Bardenpho) process for nitrification-denitrification-biological excess phosphorus removal.

In assessing the functioning of the Phoredox process, the reservation of a portion of the unaerated mass fraction to the anaerobic reactor concomitantly reduces the denitrification potential of the process. Thus, com-

pared to the Bardenpho process for the same total unaerated mass fraction, the Phoredox has a lower denitrification potential. Consequently the maximum TKN/COD ratio the Phoredox process can denitrify completely is reduced. For a "standard" raw sewage influent, i.e. 20% of the influent COD is readily biodegradable, at sludge ages of approximately 20 days, maximum unaerated mass fraction of 0,55 and an anaerobic mass fraction of 0,15, it would seem that the maximum TKN/COD ratio for complete denitrification should not exceed 0,08. For settled sewage the readily biodegradable COD is higher $\approx 25\%$ of the total influent COD and the maximum TKN/COD ratio is about 0,09. This ratio is very dependent on the fraction of readily biodegradable COD because this fraction contributes significantly to the denitrification. Consequently the values given here are relative ones only – it is important to estimate the true readily biodegradable fraction experimentally.

As the TKN/COD ratio increases from low values, so it will be found that the relative volumes of the primary and secondary anoxic reactors change accordingly. To achieve optimum denitrification response at low TKN/COD ratios the primary reactor is small and the secondary reactor large; at high TKN/COD ratios, near or at the maximum allowed, the primary reactor is large and the secondary very small.

Should the TKN/COD ratio exceed the maximum, nitrate will appear in the effluent. In the Phoredox process this nitrate is recycled in the underflow to the anaerobic reactor where it utilizes preferentially some, or all (depending on the concentration of nitrate) of the readily biodegradable COD, thereby reducing the magnitude of the COD available for sequestration by the poly-P organisms. Now the concentration of nitrate in the underflow recycle is the same as that in the effluent and 1 mg ($\text{NO}_3\text{-N}$) removes 8,6 mg readily biodegradable COD, consequently, a concentration of 5 mg/l ($\text{NO}_3\text{-N}$) in the underflow with a recycle ratio of 1:1 will remove 43 mg/l of this COD fraction which, for an influent of 500 COD/l will reduce the readily biodegradable COD from approximately 100 to 60 mg/l. The only operational means for reducing the mass of nitrate transferred to the anaerobic reactor is to reduce the underflow recycle ratio but this procedure has only limited possibilities because low recycle ratios may give rise to difficulties with the secondary settling tank – at low recycles the sludge mass builds up in the tank and may cause failure of the tank when the SVI is high, or loss of sludge in the effluent may occur from rising sludge due to denitrification in the sludge mass.

The principal difficulty with the Phoredox configuration, under high TKN/COD ratios, arises from the fact that the anaerobic reactor performance is dependent on the nitrate in the effluent. To make the anaerobic reactor independent of the effluent nitrate, the University of Cape Town (UCT) process configuration was developed, Figure (3.12). The basic denitrification configuration is the Modified Ludzack-Ettinger process (Figure 3.2). The anaerobic reactor is, like the

*The basis for proposing the anaerobic reactor, by Barnard, differs fundamentally from the hypothesis suggested in the previous sections. He suggested that the *redox potential* need to be lowered sufficiently to induce P release. He noted the adverse effect on P removal by nitrate discharged to the anaerobic reactor but suggested that this arose from the nitrate limiting the depression of the redox potential. These aspects are dealt with in greater detail in Chapter 7.

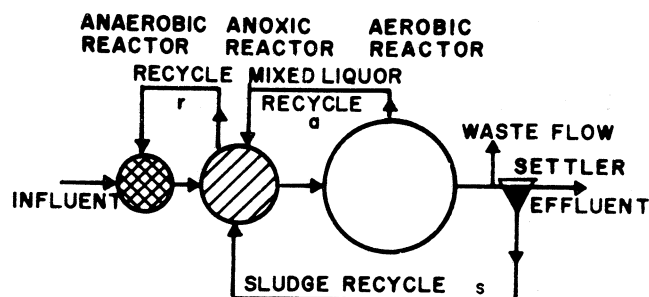


FIGURE 3.12: University of Cape Town (UCT) process for nitrification-denitrification-biological excess phosphorus removal.

Phoredox, placed at the head of the works but the recycle flow to this reactor, the r-recycle, is taken from the primary anoxic reactor. By regulating the main internal recycle from the aerobic to the anoxic reactor, the a-recycle, so that the denitrification potential of the primary anoxic reactor is not exceeded, a nitrate-free effluent is assured to the anaerobic reactor.

In the UCT process, if, say, an r-recycle of 1:1 is used the concentration in the anaerobic reactor is half that in the rest of the plant and to get the required anaerobic mass fraction the volume of this reactor must, in consequence, be increased to twice the volumetric size of that in the Phoredox for the same anaerobic mass fraction. The main advantage of the UCT process is that by operational means the anaerobic reactor can be made independent of the nitrate in the effluent, and this can be maintained up to a TKN/COD ratio of about 0,14. As an alternative to the Phoredox process, at low TKN/COD ratios, i.e. $\text{TKN/COD} < 0,08$, the UCT process is also satisfactory although the nitrate removal will not be as high as in the Phoredox as it is not possible ever to get zero nitrate in the effluent from a UCT process.

To overcome the need for explicit control of the a-recycle, the Modified UCT process was developed, (Figure 3.13). In this process the primary anoxic reactor is divided into two fractions, the first having an anoxic

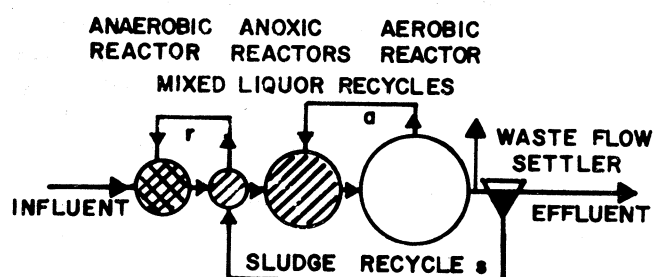


FIGURE 3.13: Modified UCT process.

mass fraction of about 0,1. The a-recycle is discharged to the second anoxic reactor and remains fixed at say 4:1 ratio irrespective of whether nitrate is recycled back to the aerobic reactor or not, provided the a-recycle ratio is so large that it brings enough nitrate to the second anoxic reactor to equal or exceed its denitrification potential. The underflow recycle (s-recycle) discharges to the first anoxic reactor, to reduce the nitrate to zero. By means of this configuration zero nitrate in the r-recycle can be maintained for TKN/COD ratios up to about 0,11.

4. CLOSURE

There are numerous aspects and details in the design of nutrient removal plants that have not been dealt with in this short review – attention was focussed on the major aspects of a fundamental nature only. For details of design reference needs to be made to Chapters 4 to 7. Designs that conform to the recommendations laid down in these chapters, for treating normal municipal wasteflows, settled or unsettled, will have a high expectation of phosphorus removal of about 1,7 to 2 mg $\text{PO}_4\text{-P}$ per 100 mg influent COD, i.e. a $\Delta\text{P/COD}$ of 0,017 to 0,02 mgP/mg influent COD. This removal ratio can vary, depending principally on the fraction of readily biodegradable COD in the influent. Should the influent P/COD concentration ratio exceed 0,017 to 0,02, very likely all the phosphorus will *not* be removed biologically from the wastewater and provision necessarily will have to be made for chemical precipitation of the remaining phosphorus; should the influent P/COD be *less* than 0,017 to 0,02, then very likely removals down to 0,5 mgP/l can be achieved for a high proportion of the time.

The above biological P removal propensity in $\Delta\text{P/COD}$ forms an approximate basis for judging the performance of any designed plant. *The plant should not be assessed on the basis of the effluent P concentration only*, but on the ΔP per unit influent COD. If the $\Delta\text{P/COD}$ achieved is less than 0,017 to 0,02, the cause for the lower removal should be sought in the wastewater characteristics, process design or operation of the plant, or in combinations of these three. The results of such investigations would be of great value to research workers and designers in the endeavour to build plants that perform optimally.

5. REFERENCES

- DOLD, P.L., EKAMA, G.A. and MARAIS, G.v.R. (1980) A general model for the activated sludge process, *Prog. Wat. Tech.*, **12**, 47-77.
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